activity, the positivists were forced to focus solely on issues in logic and the philosophy of science—pushed to what their now-distant manifesto had called “the icy slopes of logic.”

But the final blow against positivism came from within philosophy, not from the external forces of geopolitics and chance that had left it battered. New arguments against some of the central tenets of positivism came from a new generation of philosophers. These arguments laid bare the inadequacy of the verification theory of meaning and the instrumentalist account of science—and turned philosophers of science against the Copenhagen interpretation.

One of the young philosophers who had visited the Vienna Circle in its heyday was a brilliant American student with the improbable name of Willard Van Orman Quine. Quine had written a PhD thesis on mathematical logic at Harvard in 1932; for the year afterward, he traveled Europe on a fellowship, meeting with Schlick, Frank, Ayer, and other leading positivists. He spent six weeks studying with Carnap in Prague—“my first really considerable experience of being intellectually fired by a living teacher rather than by a dead book,” Quine said later. Returning from Europe “an ardent disciple of Carnap” (and with only seven dollars in his pocket), Quine went back to Harvard and taught classes on positivist philosophy. He also did important work in mathematical logic. But as Quine continued to work and teach—interrupted only by World War II, when he cracked coded messages from Nazi submarines—doubts about positivist dogma seeped in and accumulated. Finally, in 1951, the dam burst: Quine wrote a paper that brought positivism to its knees.

Quine’s paper, “Two Dogmas of Empiricism,” took aim at the core of the positivist program, the verification theory of meaning. Quine pointed out that there was no way to verify single statements—all attempts to verify a statement inevitably involve the assumed truth of other statements, which are themselves subject to the same problem. For example, say the remote control for your TV isn’t working, and you can’t turn the TV on. You suspect that the batteries in the remote are dead. You can verify this by